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1 Git and Version Control

1.1 Some motivating examples

- While working on your analysis code, you accidentally delete the first 35 lines of the script. You only discover this three days later, when you restart R and try to run the script from the top. Because of the time you last half of your senior thesis, you've been saving every couple of minutes.

- You're working on a paper with two coauthors. You prepare the final draft to send for submission: `paper final.docx`. But one of your coauthors discovers a typo. Now it's `paper final fixed typo.docx`. Another realizes six references are missing. `paper final fixed typo refs.docx`. That's getting confusing so you change it to `paper 2 Aug 2021.docx`. Once it comes back from review you need to make revisions. Now you have `paper 30 Jan 2022.docx` and, after your collaborators make their changes, `paper 12 February 2022 DJH.docx` and `paper 12 February 20222 final.docx`.
- You have a complicated analysis spread over several scripts. You want to explore a variant analysis, but doing so will involve changes in 15 different places across 3 different files. You're not sure if this variant analysis will work; you may or may not want to keep it.

1.2 Version control

- Basic idea: Tools for tracking and reversing changes to code over time
- Useful for identifying and reversing breaking changes
- Implementations upload to cloud, track who contributes code, control who can suggest vs. actually change code
- Good for collaboration, publishing code
- git
  - One of many version control systems
  - Very popular in part thanks to GitHub, which provided free hosting for open-source projects
    - In April 2020, GitHub also made private/closed-source repositories free
    - Resources for students (and teachers): https://education.github.com/

1.3 How to do things with git

- Getting help
  - `git fetch -h`
  - On Mac/Unix: `man git-fetch` (hit `q` to exit)
  - On Windows: `git help fetch`
  - A good cheat sheet: https://www.atlassian.com/git/tutorials/atlassian-git-cheatsheet
- Interfaces
  - Command line
  - RStudio projects
  - GUIs: GitHub, Sourcetree
- Setup

  - Installation
  - User config

```
$ git config --global user.name "Maria Hernandez"
$ git config --global user.email mhernandez123@ucmerced.edu
$ git config --global core.editor "nano"
```

  - You don't have to use `nano`, but it's the best option for beginners. For more on text editor options, see here https://git-scm.com/book/en/v2/Appendix-C%3A-Git-Commands-Setup-and-Config#_core_editor

- `git init`
  - Create a folder called `test` and open a terminal window there.

```
$ git status
$ git init
$ git status
```

- `add` and `commit` a file Create a text file and save it as `readme.txt` in `test`.

```
$ git status
$ git add test.txt
$ git status
$ git commit -m "my first commit"
$ git status
```

After adding a file, you can add all changes to tracked files using `-u`:

```
$ git status
$ git add -u
$ git status
$ git commit -m "-u is a shortcut"
```

NB You always need to `add` and then `commit`. Doing just one by itself doesn't do much.

- `.gitignore`
  - Create a subfolder `images` and put an image file of some kind in it.
  - Create a text file `.gitignore` in `test`. Put the following line in:

```
images
```

  - Add another image file. Edit `.gitignore` so that it ignores the first image file but not the second.
  - Track `.gitignore`

1.4 Working with GitHub remotes

- *remote*: A copy of a repository that lives on a server somewhere else

1.4.1 Working with *your own* repos

- On GitHub, click "New" and walk through the steps to create a new repository. Let's call it `mytest`.
- Copy the URL: `https://github.com/username/mytest`
- Back in the Terminal window:

```
$ git remote add origin https://github.com/username/mytest
```

You'll need to authenticate with your GitHub password. If you forget the exact incantation, try `git push`; it will give you a helpful example.

- `push` and then refresh GitHub in your browser

```
$ git push
```

- Edit the repository (eg, use the prompt near the bottom to create a readme). Then:

```
$ git fetch  ## Downloads a comparison, but doesn't change your local files yet
$ git status
$ git pull   ## Changes your local files
$ git status
```

1.4.2 Working with *someone else's* repos

GitHub lets you download someone else's repo ("clone"), and modify it locally, but *not* upload directly. You can suggest a change to someone else's code by submitting a *pull request*, which requires *forking* the repository.

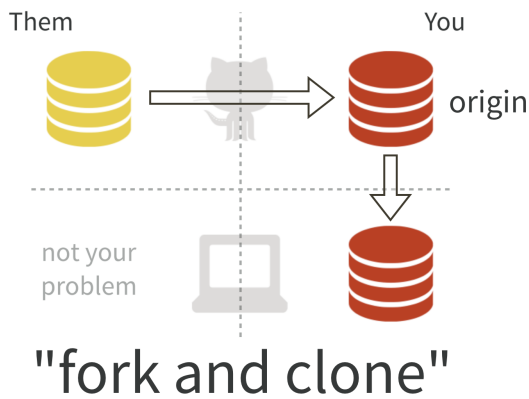

Figure 1: *Forking* copies a repository to your GitHub account. Then you *clone* the copy to your local machine. You can *push* to your remote copy as usual. You can suggest changes to the original using a *pull request*. Source: <https://happygitwithr.com/fork-and-clone.html>

- A demonstration repository: https://github.com/data-science-methods/git-demo
- *Fork step*: Look for the `fork` button in the upper-right

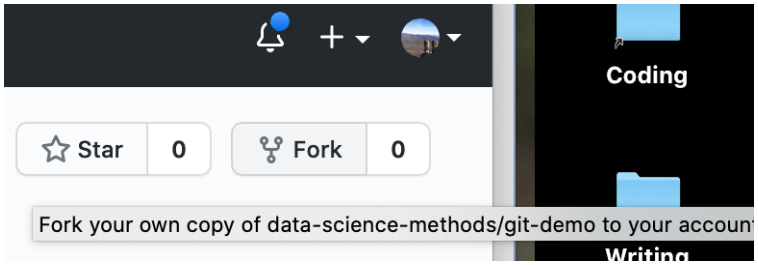

Figure 2: The *fork* button is near the upper-right corner of a GitHub repository page. I wasn't able to find a keyboard shortcut for this. :-(

- *Clone step*: After creating the fork, you need to download a copy to your machine. Open a Terminal window to the folder *right above* where you want the repository folder to be located, eg, `projects`.

```
$ git clone https://github.com/YourUsername/git-demo
$ cd git-demo
$ git status
```

- *Push step*: Make changes locally, then push them back up to *your* GitHub copy of the repo

1.5 Lab

The lab for this section can be found at https://github.com/data-science-methods/lab-w2-git.